Advanced Fluid Mechanics Course

Final Project

Boundary-Layer Flow Analysis over a Cylinder

Instructor
Dr. Amir Nejat

Student
Hamid Motallebi

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College of Engineering, University of Tehran

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Abstract

This study investigates the laminar boundary-layer flow over a stationary circular cylinder using two numerical approaches: the classical boundary-layer formulation and the full Navier–Stokes equations. In the first approach, the Prandtl boundary-layer equations are solved numerically in MATLAB using an implicit finite-difference marching scheme. Since the numerical solution cannot be initiated directly at the stagnation point, a Falkner–Skan similarity solution is employed to generate the initial velocity profile. The resulting system of discretized equations is solved using the Thomas algorithm for tridiagonal matrices. Velocity profiles, boundary-layer thickness, displacement thickness, momentum thickness, and the separation location are subsequently determined at different angular positions around the cylinder. The MATLAB solution predicts boundary-layer separation at approximately **87.57°**, and the calculated velocity profiles show good qualitative agreement with the reference profiles presented in White's *Fluid Mechanics*.

In the second approach, the complete laminar flow field around the cylinder is simulated using ANSYS Fluent at a Reynolds number of approximately **9500**. Velocity and pressure contours, velocity vectors, streamlines, and boundary-layer characteristics are extracted from the CFD results. The CFD simulation indicates flow separation at approximately **80°**, earlier than that predicted by the boundary-layer solution. Comparison of the two approaches demonstrates close agreement in the attached-flow region, while the discrepancies increase as the adverse pressure gradient becomes stronger and the flow approaches separation. In particular, the full Navier–Stokes solution predicts a more rapid growth of the boundary layer and is capable of representing flow reversal and the separated wake, whereas the classical boundary-layer equations lose their validity after separation. The results therefore illustrate both the computational efficiency of boundary-layer theory and its limitations in predicting strongly separated flows.

Keywords: Boundary Layer; Circular Cylinder; Flow Separation; Navier–Stokes Equations; Computational Fluid Dynamics (CFD); Finite Difference Method; MATLAB; ANSYS Fluent.

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Introduction

The study of fluid flow over a circular cylinder is one of the classical and fundamental problems in fluid dynamics. Owing to its broad engineering applications and its physically rich behavior, it has consistently attracted the attention of researchers. Applications range from airflow over electrical transmission lines, chimneys, and bridge piers to water flow in shell-and-tube heat exchangers. A precise understanding of aerodynamic forces, pressure distribution, and especially flow separation is essential for the safe and optimal design of such structures. This problem also serves as a standard benchmark for evaluating and validating analytical, experimental, and numerical methods in fluid mechanics.

At the heart of this flow physics is the boundary-layer concept, first introduced by Ludwig Prandtl. According to this theory, at high Reynolds numbers the effects of fluid viscosity are confined to a thin region adjacent to the body surface, while the flow outside this layer can be approximated as inviscid with good accuracy. This useful decomposition greatly simplifies the flow analysis. On the front portion of the cylinder, the accelerating flow produces a favorable pressure gradient, and the boundary layer remains attached to the surface while it develops. As the flow passes over the upper portion of the cylinder and moves toward the rear, however, an adverse pressure gradient (an increase in pressure in the direction of flow) develops. This opposing pressure gradient reduces the kinetic energy of fluid particles near the wall and ultimately leads to one of the most complex phenomena in fluid dynamics: flow separation (Flow Separation). At the separation point, the wall shear stress becomes zero, the flow detaches from the surface, and a broad, unsteady wake (Wake) forms behind the cylinder; this wake is a primary source of drag (Drag).

Two principal approaches, with different levels of accuracy and computational cost, are available for analyzing this complex flow:

1- Boundary-Layer Equation Solution (Boundary Layer Theory): This classical approach is based on solving the simplified Navier-Stokes equations (Prandtl boundary-layer equations). Because these equations are parabolic (Parabolic), they can be solved by marching step by step in the flow direction and are computationally very efficient. In this project, the equations are solved using an implicit finite-difference numerical method following the procedure given in White's reference textbook, and the corresponding code is implemented in MATLAB. The main limitation of this method is that, after the separation point, the appearance of reverse flow invalidates the governing boundary-layer equations and the solution must be terminated.

2- Full Navier-Stokes Solution (Navier-Stokes Solution): In this approach, now commonly implemented using computational fluid dynamics (CFD), the complete equations governing fluid motion are solved throughout the entire domain without the simplifying assumptions of boundary-layer theory. It can simulate complex flow details, including the precise separation point, wake-vortex structure, and interaction between the boundary layer and the external flow, with high

fidelity. Although computationally more expensive, this approach provides a more complete and realistic representation of the flow physics.

The main objective of this project is to implement both methods described above for flow over a cylinder at Reynolds number $Re = 9500$ and then perform a detailed comparative analysis of their results. The key boundary-layer parameters considered are the velocity profile, boundary-layer thickness (δ), displacement thickness (δ^*), and momentum thickness (θ). These quantities are calculated and compared at different angular positions. The comparison focuses on understanding and explaining the physical and mathematical reasons for the differences between the two solutions, especially near separation, so that the strengths and limitations of the two analytical approaches can be clearly identified.

This report is organized into two main parts. Part (A) presents the theoretical background, the finite-difference numerical solution method, and the results obtained from the MATLAB code for solving the boundary-layer equations. Part (B) describes the CFD simulation procedure, software settings, and results from the full Navier-Stokes solution. Finally, the results of the two parts are compared directly to provide a comprehensive analysis of the flow behavior and to evaluate the accuracy of the boundary-layer model in predicting this classical phenomenon.

Part (A): Solution of the Boundary-Layer Equations over a Stationary Cylinder

3-1 Theoretical Background and Governing Equations

To solve the boundary-layer flow over the cylinder, the boundary-layer form of the Navier-Stokes equations is used. Because the solution starts at the stagnation point $(Stagnation\ Point)$ where $x = 0$, the finite-difference method cannot be applied directly. Therefore, a similarity solution must first be used.

3-2 Governing Equations and Similarity Solution $(Falkner - Skan)$

The similarity solution of the Falkner-Skan equation provides the following results for initializing the flow:

$$u = U_\infty f'(\eta)$$

where the similarity variable η and parameter B are defined as follows (for the cylinder stagnation point, $m = 1$):

$$\eta = y\sqrt{\frac{m+1}{2}\frac{U_\infty}{\nu x}}, \qquad B = \frac{2m}{m+1}$$

The governing ordinary differential equation and its boundary conditions are:

$$\{f''' + ff'' + B(1 - f'2) = 0$$
$$f(0) = f'(0) = 0,$$
$$f'(\infty) = 1$$

The numerical solutions of this equation (the profile $f'(\eta)$) are used as the initial velocity profile at the first computational node ($i = 1$) in the finite-difference code.

3-3 Displacement Thickness (Definitions and Concepts)

Writing conservation of mass for the control volume in Figure 2 gives Equation 8.

$$\iint_{CS} \rho V.dA = 0 = \int_0^Y \rho u dy - \int_0^H \rho U dy$$

Assuming incompressible flow, Equation 8 reduces to Equation 9.

$$UH = \int_0^Y u dy = \int_0^Y (U + u - U)dy = UY + \int_0^Y (u - U)dy$$

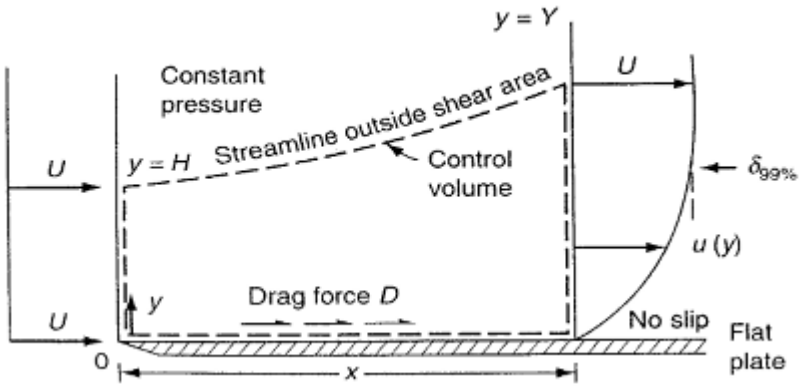


*Figure*1 *: Displacement Thickness and Momentum Thickness*

Rearranging Equation 11 and defining $Y = H + \delta^*$ gives the mass-flow relation in Equation 10.

$$U(Y - H) = U\delta^* = \int_0^Y (U - u)dy \;\Rightarrow\; \delta^* = \int_0^{Y\to\infty} \left(1 - \frac{u}{U}\right) dy$$

The above equation is the formal definition of displacement thickness and is valid for incompressible laminar or turbulent flow, with constant or variable pressure and constant or variable temperature.

In this project, for the first two cases, the integral relation in the above equation is evaluated using the MATLAB code to obtain the displacement thickness. In Fluent, the integral is likewise evaluated along the defined line at each angle to obtain the corresponding results.

3-4 Momentum Thickness

The momentum thickness is expressed by the following relation.

$$\theta = \int_0^{Y\to\infty} \frac{u}{U}\left(1 - \frac{u}{U}\right) dy$$

3-5 Use of the Empirical Correlation

The Hiemenz experiment for a cylinder at Reynolds number $Re_a = U_\infty a/\upsilon = 9500$, where a is the cylinder radius, shows that the polynomial in Equation 13 provides a suitable fit to these results.

$$\frac{U}{U_\infty} \approx 1.814\, x^* - 0.271 x^{*3} - 0.0471 x^{*5} \ldots$$

This differs completely from potential flow. This relation predicts separation at an angle of $\varphi = 80.5°$.

3-6 Geometry and Grid Generation

The problem geometry is represented by a rectangular computational domain (x along the cylinder surface and y normal to it).

- Inlet boundary (left): Velocity profile obtained from the Falkner-Skan solution.
- Lower boundary: no-slip condition ($u = 0, v = 0$).
- Upper boundary: free-stream velocity $U_e(x)$

For potential flow over a cylinder, the standard relation is $U_e = 2U_\infty \sin(x/R)$. In the code below, the standard theoretical (sinusoidal) relation is used to remain consistent with the results in White's textbook at ($Re = 9500$); however, the polynomial relation is also included in the code comments.

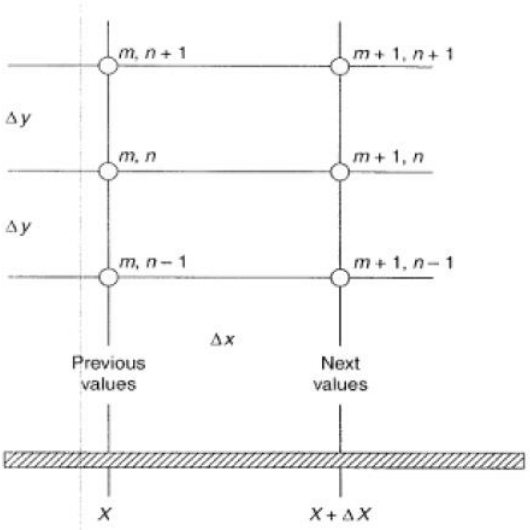


Figure 2:Finite-Difference Grid for a Two-Dimensional Boundary Layer

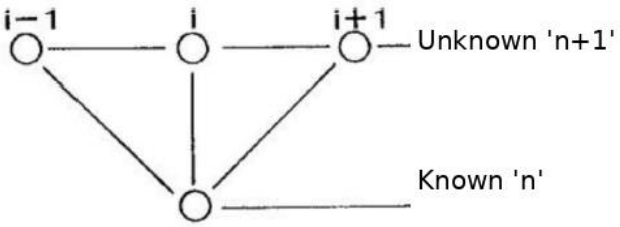


Figure 3: Grid Points for the Implicit Formulation

3-7 Numerical Solution Method: Implicit Finite Difference

To march along the cylinder ($Marching$) an implicit method is used. In this method, derivatives in the x direction are written using a backward difference ($Backward$) and derivatives in the y direction using a central difference ($Central$) at the new location $i + 1$.

The discretized momentum equation for the interior grid points is written as follows:

$$-\alpha u_{i+1,j+1} + (1 + 2\alpha)u_{i+1,j} - \alpha u_{i+1,j-1} = u_{i,j} - \beta\left(u_{i,j+1} - u_{i,j-1}\right) + \frac{U_{e,i+1}^2 - U_{e,i}^2}{2U_{i,j}}$$

where the coefficients α and β depend on the grid steps Δx,Δy and the velocities from the previous step $u_{i,j}, v_{i,j}$.

This equation produces a tridiagonal matrix ($Tridiagonal\ Matrix$).

$$\begin{bmatrix} 1+2\alpha & -\alpha & 0 & \cdots & 0 & 0 \\ -\alpha & 1+2\alpha & -\alpha & \ddots & 0 & 0 \\ \vdots & 0 & 0 & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \vdots & -\alpha & 1+2\alpha \end{bmatrix} [u] = [d]$$

This system of equations is solved at each spatial step x using the Thomas algorithm ($TDMA$).

3-7-1 Flow over a Cylinder

The problem considered is incompressible viscous flow over a circular cylinder. The boundary layer begins at the stagnation point ($Stagnation\ Point$) at $\theta = 0$ and grows with increasing angle until the adverse pressure gradient causes separation ($Separation$).

According to Prandtl boundary-layer theory, the governing equations are:

1- **Continuity equation:**

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

2- **Momentum equation in the x direction:**

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = U_e\frac{dU_e}{dx} + \nu\frac{\partial^2 u}{\partial y^2}$$

Here:

- x: coordinate along the curved surface of the cylinder.
- y: coordinate normal to the surface.
- $U_e(x)$ is the free-stream velocity at the edge of the boundary layer ($Edge\ Velocity$).

3-7-2 Free-Stream Velocity Distribution (*Hiemenz Flow*)

For a real cylinder, the exact potential-flow velocity is not $2U_\infty \sin(\theta)$ because of viscous and wake effects. The more accurate Hiemenz empirical relation for the dimensionless velocity distribution is given below and is used in the code:

$$U_e(x) \approx U_0(1.814\phi - 0.271\phi^3 - 0.0471\phi^5)$$

where $\phi = x/R$ is the angle in radians and U_0 is the inlet-flow velocity. The Reynolds number for the problem is $Re_R = \frac{U_0 R}{\nu} = 9500$, as specified.

3-7-3 Implicit Numerical Solution Method

According to pages 275 and 276 of White's textbook, the explicit method $(Explicit)$ has severe stability restrictions ($Eq\ 4-147$). White therefore recommends the implicit method ($Implicit$) which is unconditionally stable.

3-7-4 Discretization

A rectangular computational grid (m, n) is considered, where:

- m: index in the x direction (flow direction).
- n: index in the y direction (normal to the wall).

In the implicit method, the second derivative with respect to y (the viscous term) is written at the new spatial step $(m+1)$. This couples the unknowns in a system of equations.

Discretized momentum equation (Equation 4-149 in White's textbook):

We begin the solution by writing a finite-difference model for the momentum equation corresponding to row n of the grid.

$$u_{m,n}\frac{u_{m+1,n}-u_{m,n}}{\Delta x}+v_{m,n}\frac{u_{m,n+1}-u_{m,n-1}}{2\Delta y} \approx \frac{U_{m+1}^2-U_m^2}{2\Delta x}+\upsilon\frac{u_{m,n+1}-2u_{m,n}+u_{m,n-1}}{\Delta y^2}$$

Starting from the left, each term is examined in turn. The first term uses a forward difference to model $\partial u/\partial x$. The second term uses a central difference to represent $\partial u/\partial y$. This term is also written at row n. The third term is the pressure-gradient term. In this term, the relation $U(dU/dx)=d(U^2/2)$ is first used, followed by a forward difference. Note that U_{m+1} is known because U(x) represents the free-stream velocity.

The fourth and final term uses a central-difference approximation for $\partial^2 u/\partial y^2$. In all terms, row n is at the center of the stencils, and the stencils are centered with respect to it, which is desirable for numerical computation. The only unknown velocity in this stencil is $u_{m+1,n}$, so the equation can be solved for it to obtain the following result:

$$u_{m+1,n}\approx(\alpha-\beta)u_{m,n+1}+(1-2\alpha)u_{m,n}+(\alpha+\beta)u_{m,n-1}+\frac{U_{m+1}^2-U_m^2}{2u_{m,n}}$$

where:

$$\alpha=\frac{\upsilon\Delta x}{u_{m,n}\Delta y^2} \quad , \quad \beta=\frac{v_{m,n}\Delta x}{2u_{m,n}\Delta y}$$

All terms on the right-hand side of the above equation are known from the previous step. Therefore, $u_{m+1,n}$ can be readily obtained; this is referred to as the explicit method. For stability in this case, $\alpha<0.5$ and $\beta<\alpha$. This requirement restricts the grid-step sizes in both directions.

$$\Delta x<\frac{u_{min}\Delta y^2}{2\upsilon} \quad , \quad \Delta y<\frac{2\upsilon}{|v_{max}|}$$

To impose a bound on Δy we need v_{max} because v is positive under an adverse pressure gradient but may be negative under a favorable pressure gradient. Usually u_{min} occurs near the wall and v_{max} near the free stream. During these explicit calculations, $\alpha(y)$ and $\beta(y)$ must be monitored to ensure that the conditions above remain satisfied.

Once $u_{m+1,n}$ is determined from Equation 4, $v_{m+1,n}$ is obtained by discretizing the continuity equation (Equation 1). The simplest form, using a forward difference for both terms, is:

$$\frac{u_{m+1,n} - u_{m,n}}{\Delta x} + \frac{v_{m+1,n} - v_{m+1,n-1}}{\Delta y} \approx 0$$

We now start at n = 2 and calculate $v_{m+1,2}$ because $v_{m+1,n} = v_w$ is known. The first term is written at row n, whereas the second term is at row n - 1/2. The resulting numerical accuracy is low. Writing $\partial v/\partial y$ as a central difference does not resolve the issue either. One proposed remedy is the following relation:

$$\frac{\partial u}{\partial x}_{avg} = 1/2\left(\frac{u_{m+1,n} - u_{m,n}}{\Delta x} + \frac{u_{m+1,n-1} - u_{m,n-1}}{\Delta x}\right)$$

The vertical velocity at the next step is also obtained as follows:

$$v_{m+1,n} \approx v_{m+1,n-1} - \frac{\Delta y}{2\Delta x}[u_{m+1,n} - u_{m,n} + u_{m+1,n-1} - u_{m,n-1}]$$

This relation does not introduce instability unless the conditions of Relations 5 stated previously are violated. The solution is started at n = 2 and marched upward.

The equations above provide a satisfactory explicit laminar boundary-layer model. During the calculations, the stability condition must be monitored continuously. The values at the outer boundary $u_{m,N-1}$ and $u_{m,N}$ should also be checked to verify that Um converges toward the free-stream velocity.

3-7-5 Solution of the Tridiagonal Matrix ($TDMA$)

The above system is written for all points from $n = 2$ to $N - 1$.

- Lower boundary condition (wall): $u_{m+1,1} = 0$
- Upper boundary condition (free stream): $u_{m+1,N} = U_{e,m+1}$

This linear system is solved using the Thomas algorithm ($Thomas\ Algorithm$) which is very fast.

In the implicit method (Implicit), after discretizing the differential equation, the equation at each node j in the vertical direction takes the following form:

$$a_j u_{j-1} + b_j u_j + c_j u_{j+1} = d_j$$

where:

- u_j: unknown velocity at the current node.
- u_{j-1} : unknown velocity at the lower node.
- u_{j+1} : unknown velocity at the upper node.
- a_j, b_j, c_j : known coefficients (dependent on viscosity, spatial step, and velocities from the previous step).
- d_j : right-hand-side quantity (RHS) containing values from the previous time or spatial step.

When these equations for all nodes ($j = 1$ through N) are assembled, the coefficient matrix contains nonzero values only on the main diagonal, the upper diagonal, and the lower diagonal; all other entries are zero:

$$\begin{bmatrix} b_1 & c_1 & 0 & \cdots & 0 & 0 \\ a_2 & b_2 & c_2 & \ddots & 0 & 0 \\ \vdots & a_3 & b_3 & \ddots & \vdots & c_{N-1} \\ 0 & 0 & \dots & \vdots & a_n & b_n \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_N \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ \vdots \\ d_N \end{bmatrix}$$

3-7-6 Steps of the Thomas Algorithm ($TDMA$)

This algorithm consists of two main stages:

1- Forward elimination ($Forward\ Elimination$): eliminate the lower diagonal (a) and modify the coefficients.

2- Back substitution ($Backward\ Substitution$): determine the unknowns from bottom to top.

Step 1: Forward Elimination

The objective is to eliminate the coefficients a_i so that the matrix becomes upper triangular. The coefficients c and d are modified (denoted by c' and d').

For the first row ($i = 1$):

$$c_1' = \frac{c_1}{b_1}$$

$$d_1' = \frac{d_1}{b_1}$$

For the subsequent rows ($i = 2, \dots, N$):

$$c_i' = \frac{c_i}{b_i - a_i c_{i-1}'}$$

$$d_i' = \frac{d_i - a_i d_{i-1}'}{b_i - a_i c_{i-1}'}$$

The denominator $(b_i - a_i c_{i-1}')$ must not become zero; in CFD this generally does not occur because the main diagonal is dominant.

Step 2: Backward Substitution

After reaching the last row, the final unknown (Nu_N) can be determined directly, after which the solution proceeds backward.

For the last node $(i = N)$

$$u_N = d'_N$$

For the remaining nodes ($i = N-1, N-2, \dots, 1$):

$$u_i = d'_i - c'_i u_{i+1}$$

$$-A_j u_{j-1} + B_j u_j - C_j u_{j+1} = D_j$$

Boundary conditions:

1. At the wall $(j = 1)$: the velocity is zero $(u_1 = 0)$.

- Therefore, in the matrix:$b_1 = 1, c_1 = 0, d_1 = 0$

2. At the free stream $(j = N)$ the velocity is equal to U_e; hence $(u_N = U_e)$

- Therefore, in the matrix: $a_N = 0, b_N = 1, d_N = U_e$

3-7-7 MATLAB Programming

This code was written with high accuracy while observing all of the points discussed above, including stability near the wall and accurate calculation of v .

Code features:

1. Fine grid: To achieve high accuracy near the separation point, an angular step of $d\theta = 0.02°$

is used.

2. **Calculation of v**: the vertical velocity v is calculated at each step by integrating the continuity equation (required for coefficient β).
3. **Separation detection: the separation condition** ($\tau_w \leq 0$) is checked carefully.

3-7-8 1-8-1 Code Outputs

A sample of the code outputs is presented first; all outputs are then listed in the following pages:

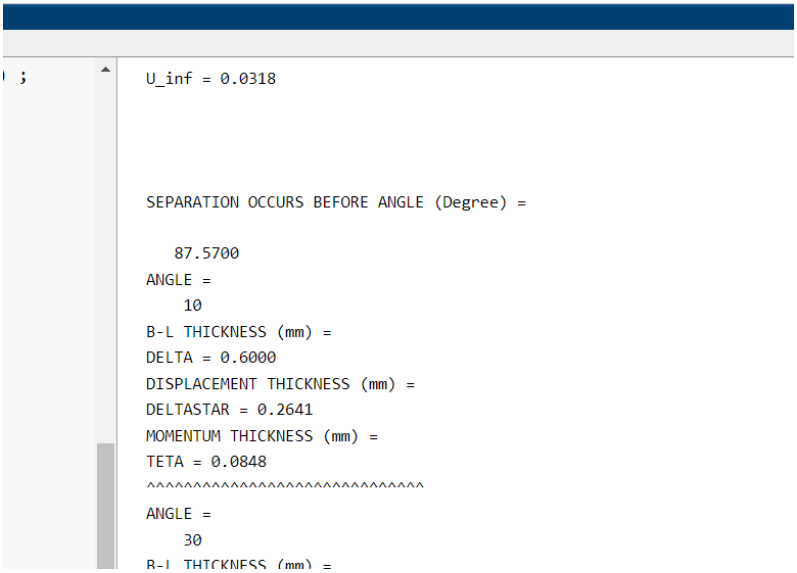


Figure 4: Sample Code Output for Flow over a Stationary Cylinder

The remaining results obtained from the code are as follows:

3-7-9 Output Values of Momentum Thickness, Displacement Thickness, and Boundary-Layer Thickness

U_inf = 0.0318
SEPARATION OCCURS BEFORE ANGLE (Degree) =
87.5700
ANGLE =
10
B-L THICKNESS (mm) =
DELTA = 0.6000
DISPLACEMENT THICKNESS (mm) =
DELTASTAR = 0.2641
MOMENTUM THICKNESS (mm) =
TETA = 0.0848
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
ANGLE =
30
B-L THICKNESS (mm) =
DELTA = 1.9500
DISPLACEMENT THICKNESS (mm) =
DELTASTAR = 0.8741
MOMENTUM THICKNESS (mm) =
TETA = 0.3162
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
ANGLE =
50
B-L THICKNESS (mm) =

DELTA = 3.0750
DISPLACEMENT THICKNESS (mm) =
DELTASTAR = 1.2858
MOMENTUM THICKNESS (mm) =
TETA = 0.4888
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

ANGLE =
70
B-L THICKNESS (mm) =
DELTA = 4.1250
DISPLACEMENT THICKNESS (mm) =
DELTASTAR = 1.7790
MOMENTUM THICKNESS (mm) =
TETA = 0.6571
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

ANGLE =
75
B-L THICKNESS (mm) =
DELTA = 4.4250
DISPLACEMENT THICKNESS (mm) =
DELTASTAR = 1.9863
MOMENTUM THICKNESS (mm) =
TETA = 0.7085
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

ANGLE =
80
B-L THICKNESS (mm) =
DELTA = 4.8750
DISPLACEMENT THICKNESS (mm) =
DELTASTAR = 2.2876
MOMENTUM THICKNESS (mm) =
TETA = 0.7665
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

ANGLE =
87.5700
B-L THICKNESS (mm) =
DELTA = 5.9250
DISPLACEMENT THICKNESS (mm) =
DELTASTAR = 3.2945
MOMENTUM THICKNESS (mm) =
TETA = 0.8343
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

ANGLE =

88.7200
B-L THICKNESS (mm) =
DELTA = 6.3000
DISPLACEMENT THICKNESS (mm) =
DELTASTAR = 3.7280
MOMENTUM THICKNESS (mm) =
TETA = 0.7996
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

Accordingly, if the code outputs are summarized in a table, they are as follows:

Table 1: Displacement Thickness, Momentum Thickness, and Boundary-Layer Thickness Obtained from the MATLAB Numerical Solution

Angle (deg)	Boundary Layer Thickness δ(mm)	Displacement Thickness δ^*(mm)	Momentum Thickness θ(mm)	Status
10	0.6000	0.2641	0.0848	Normal
30	1.9500	0.8741	0.3162	Normal
50	3.0750	1.2858	0.4888	Normal
60	4.1250	1.7790	0.6571	Normal
70	4.4250	1.9863	0.7085	Normal
80	4.8750	2.2876	0.7665	Normal
87.5700	5.9250	3.2945	0.8343	Separation
88.7200	6.3000	3.7280	0.7996	Post – Separation

3-7-10 Separation Point

In the code, consistent with the explanation at the beginning of the report, the separation point is defined as the point at which, because the velocity at the wall (node $j = 1$) is zero ($u_1 = 0$) it is sufficient to check the velocity at the first node adjacent to the wall (node $j = 2$).
If the velocity at the second node (u_2) becomes negative or very close to zero, separation has occurred.

```
U_inf = 0.0318

SEPARATION OCCURS BEFORE ANGLE (Degree) =

   87.5700
ANGLE =
    10
B-L THICKNESS (mm) =
DELTA = 0.6000
DISPLACEMENT THICKNESS (mm) =
DELTASTAR = 0.2641
MOMENTUM THICKNESS (mm) =
TETA = 0.0848
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
ANGLE =
    30
```

Figure 5: Separation Point in the MATLAB Code Output

3-7-11 Plotting the Velocity Profile

After reporting the boundary-layer thickness, displacement thickness, and momentum thickness, several velocity profiles are plotted. Note that the grid subdivision used in the code is based on 5° intervals; however, to make the plotted curves clear, 20° intervals are used when displaying the velocity profiles.

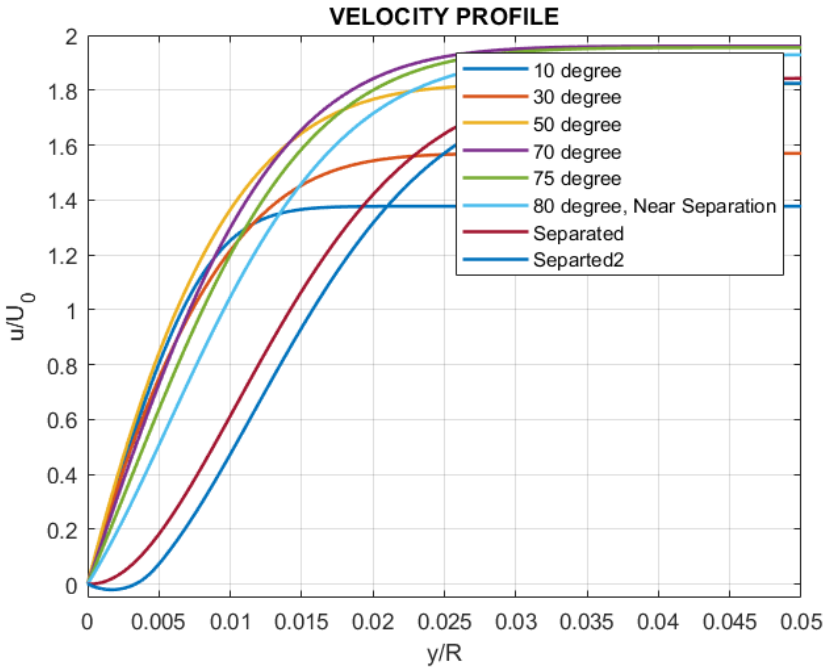


Figure 6: Velocity Profiles over a Stationary Cylinder from 0 Degrees (Stagnation Point) to Beyond the Separation Point

3-7-12 Validation of the Obtained Results

Because Part (A) of the project requires the calculated velocity profiles to be compared with Figure 4-27 of White's textbook for validation, the two figures are examined below.

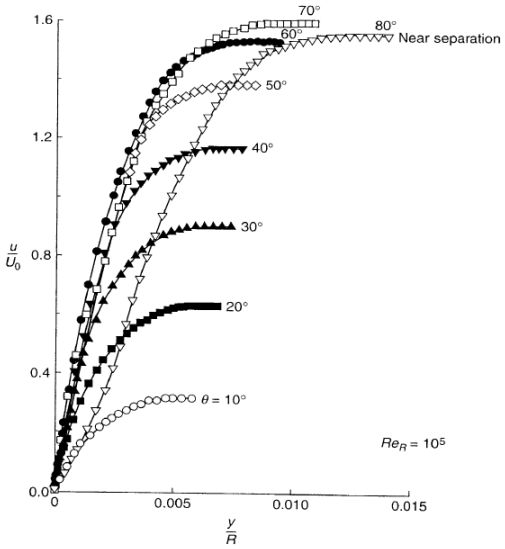


FIGURE 4-27
Laminar boundary layers on a cylinder computed by the implicit numerical method of Eqs. (4-148) and (4-149). Freestream velocity given by Eq. (4-144).

Figure 7: Figure 4-27 from White's Textbook, Used to Validate the MATLAB Numerical Results

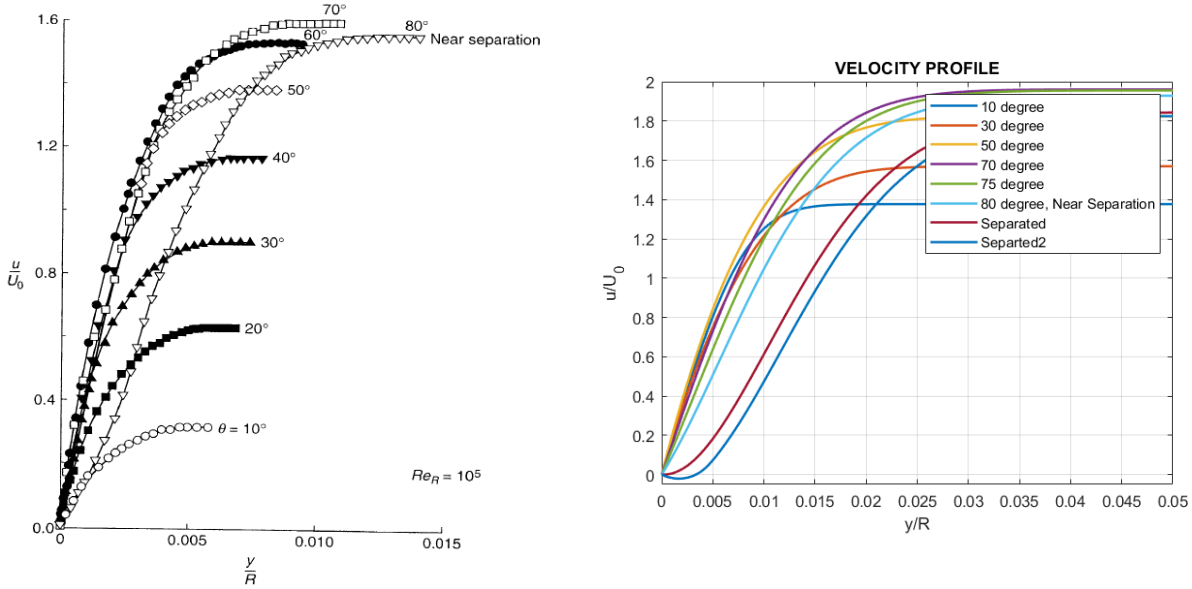


Figure8：Comparison of White's Figure 4-27 and the MATLAB Output Based on the Boundary-Layer Velocity Profile

As can be seen, the velocity profile obtained from the MATLAB numerical solution is fully consistent with Figure 4-27 and exhibits the same behavioral pattern. Therefore, the obtained results are considered reliable.

Part (B): Numerical Solution Using ANSYS Fluent

4-1 Answer to Question 1: Simulation Setup and Procedure

In this project, the boundary-layer flow around the cylinder is solved using the commercial ANSYS Fluent software. The following figures show the selected solution settings and the generated mesh (grid):

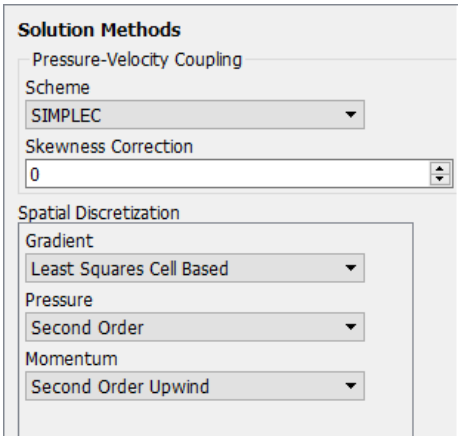


Figure 9: Fluent Solution-Method Settings

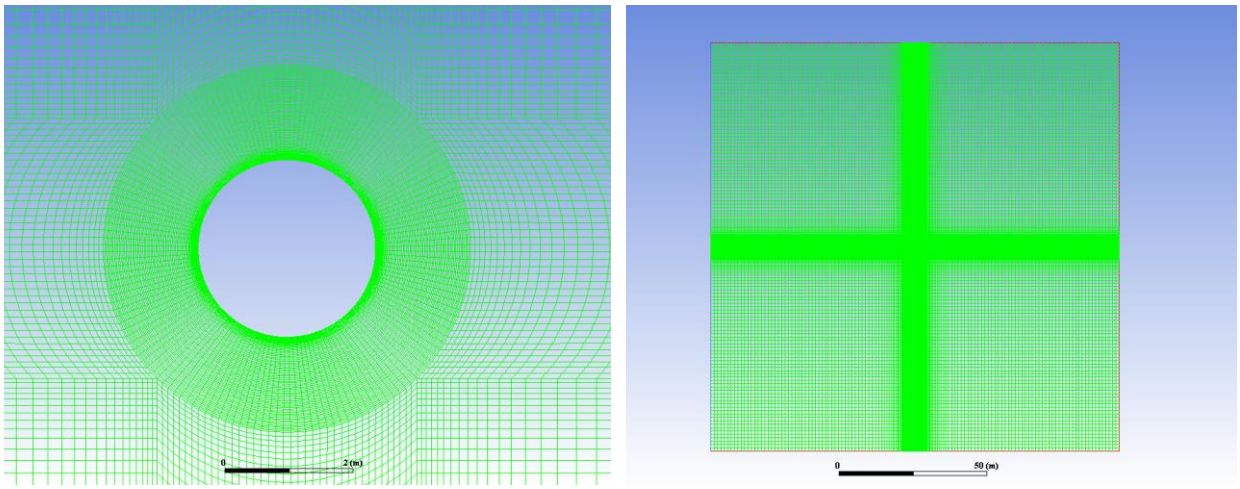


Figure10 : Mesh Generated for the Fluent Simulation

For the laminar-flow simulation over the cylinder, the velocity is set to 1 m/s. The fluid properties and cylinder radius used in the simulation are summarized in the following table:

Table2 : Fluid Properties

Density	Dynamic Viscosity	Cylinder Radius
1 kg/m^3	0.00021 $kg/(m.s)$	**1 m**

Based on the table above and the specified velocity, the Reynolds number is calculated as 9500, indicating that the flow over the cylinder is laminar.

4-1-1 Numerical Solution Results (CFD)

Using Fluent, the velocity and pressure contours for laminar flow over the cylinder are obtained as shown in the following figures:

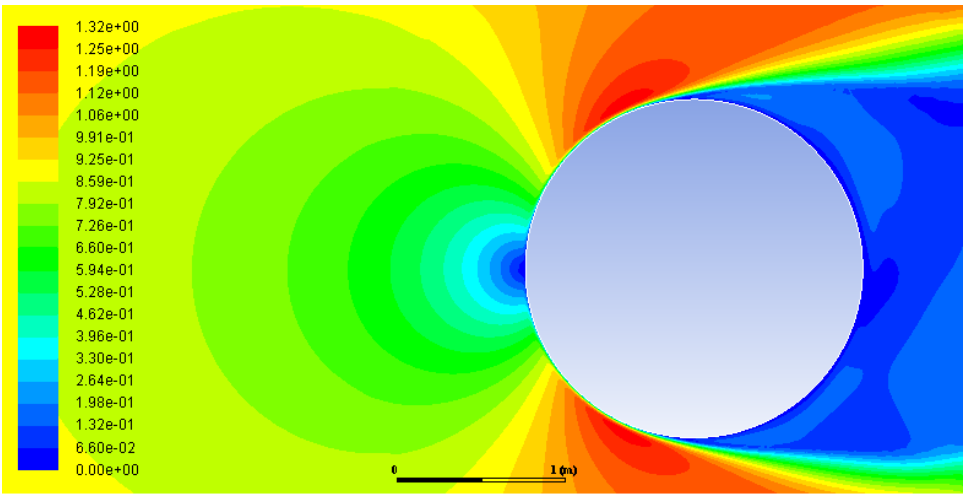


Figure 11: Velocity Contours of the Fluid Flow around the Cylinder

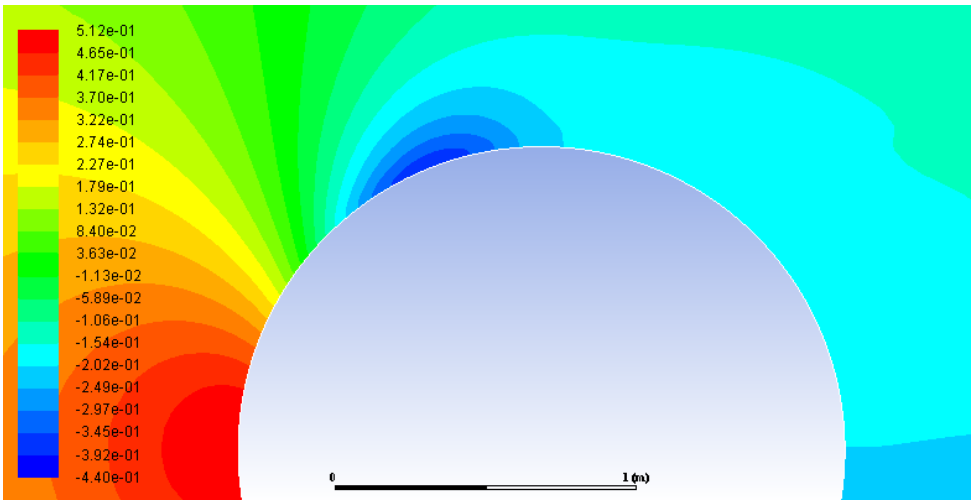


Figure 12: Pressure Contours of the Fluid Flow around the Cylinder

To determine the separation point for laminar flow around the cylinder, velocity vectors are plotted over the cylinder at 5-degree intervals. The velocity vectors indicate that the flow separates at approximately 80 degrees.

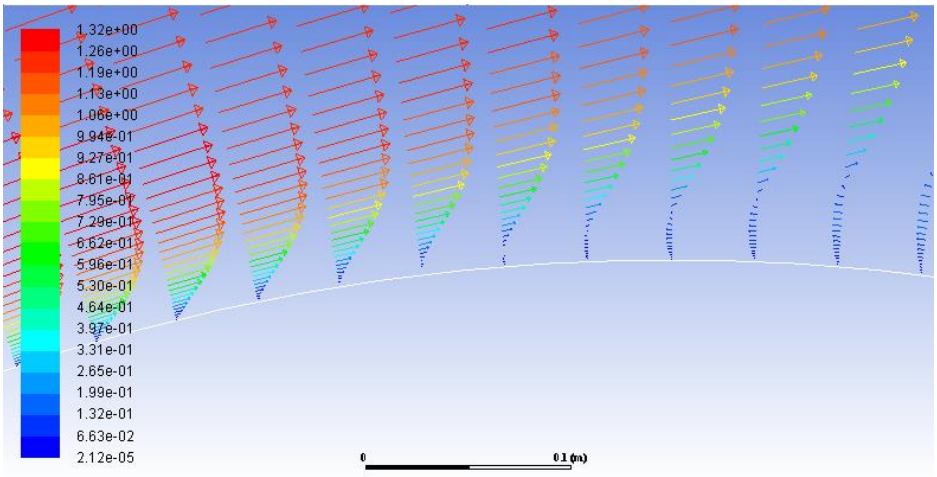


Figure 13: Fluid-Velocity Vectors around the Cylinder near the Separation Point

Next, at angles of 5, 15, ..., 75, and 80 degrees, the displacement thickness, boundary-layer thickness, and momentum thickness are calculated.

Velocity values at each angle are exported from Fluent and then organized in Excel. Finally, MATLAB is used to evaluate the required integrals numerically by the trapezoidal rule using the trapz command.

AY	AZ	BA	BB	BC
theta = 60				
r(m)	u(m/s)	u/U	1-u/U	u/U(1-u/U)
0.869304	0.990666	0.788324	0.211676	0.166721
0.859281	0.991846	0.789242	0.210758	0.166168
0.840714	0.993228	0.790321	0.209679	0.165519
0.822148	0.994675	0.791452	0.208548	0.164835
0.80358	0.996221	0.792661	0.207339	0.164101
0.785014	0.997884	0.793963	0.206037	0.163307
0.766447	0.999687	0.795377	0.204623	0.162443
0.74789	1.001643	0.796913	0.203087	0.161501
0.67362	1.008271	0.802255	0.197745	0.158396
0.655059	1.01096	0.804383	0.195617	0.157087
0.636493	1.013777	0.806612	0.193388	0.155707
0.617927	1.016734	0.808953	0.191047	0.154246
0.599362	1.019838	0.811409	0.188591	0.152701
0.580796	1.023095	0.813986	0.186014	0.151065
0.562231	1.026516	0.816693	0.183307	0.149332
0.543666	1.030107	0.819535	0.180465	0.147495
0.5251	1.033887	0.822525	0.177475	0.145543
0.506534	1.03787	0.825677	0.174323	0.143466

F	G	H	I	J
theta = 10				
r(m)	u	u/U	1-u/U	u/U(1-u/U)
0.999994429	0.668011	1	0	0
0.998676709	0.667725	0.999569601	0.000430399	0.0004302
0.980109967	0.6637275	0.993568157	0.006431843	0.0063897
0.961543383	0.659692	0.987509391	0.012490609	0.0123319
0.942976641	0.6555505	0.981291216	0.018708784	0.0183528
0.924410057	0.651302	0.974911998	0.025088002	0.0244477
0.905843315	0.646953	0.968381525	0.031618475	0.0306012
0.88727673	0.642499	0.961693074	0.038306926	0.0368135
0.868709989	0.637934	0.954837589	0.045162411	0.043086
0.850138228	0.633254	0.947809079	0.052190921	0.0494172
0.850138228	0.633254	0.947809079	0.052190921	0.0494172
0.831576663	0.6284535	0.940599257	0.059400743	0.0558068
0.813004902	0.6235285	0.933202131	0.066797869	0.0622518
0.794443336	0.6184745	0.925610875	0.074389125	0.0687494
0.775871575	0.613286	0.917817271	0.082182729	0.0752973
0.75731001	0.607958	0.909813725	0.090186275	0.0818919
0.738738249	0.602486	0.901593449	0.098406551	0.0885282
0.720176684	0.596864	0.89314742	0.10685258	0.095202
0.701604923	0.5910875	0.884468781	0.115531219	0.1019066
0.683038181	0.5851505	0.87554851	0.12445149	0.1086362
0.664471597	0.579047	0.866377519	0.133622481	0.1153837
0.645904855	0.572773	0.856949715	0.143050285	0.122139

Figure14 : Sample Data Exported from Fluent to Excel

4-2 Answer to Question 2: Comparison of Velocity Profiles between the Numerical Boundary-Layer Solution and the Navier-Stokes Solution

The main difference between the Navier-Stokes solution (CFD) and the boundary-layer solution lies in their assumptions. The full Navier-Stokes equations (NS) account for the complete flow physics, including pressure variations normal to the surface ($\partial P/\partial y \neq 0$) and surface-curvature effects, and they model the two-way interaction between the boundary layer and the external flow. In contrast, classical boundary-layer theory assumes constant pressure across the boundary layer ($\partial P/\partial y = 0$) and provides a one-way solution (the external flow affects the boundary layer, but not vice versa). These fundamental differences are the main reason for the discrepancies between the results.

Comparison at Different Angles

1- Angles of 10 and 30 Degrees (Favorable Pressure-Gradient Region):

- Profile shape: At these angles, the flow is accelerating. Both solutions show full velocity profiles that remain completely attached to the surface. The overall profile shapes are expected to be very similar.
- Reason for the difference: The discrepancy is very small. Because the NS solution fully accounts for curvature effects and interaction with the external flow, it may show a profile that reaches the free-stream velocity slightly faster. In other words, the boundary layer in the NS solution may be slightly thinner and less distorted. In practice, however, agreement between the two solutions in this region is very good.

2- Angle of 60 Degrees (Near the Minimum-Pressure Location):

- Profile shape: The adverse pressure gradient has begun to develop. The velocity profile is no longer as full as at the initial angles, and its near-wall slope (which represents wall shear stress) has decreased.
- Reason for the difference: The discrepancies become somewhat more pronounced. The NS solution models the adverse pressure-gradient effect more accurately. This causes the velocity profile in the NS solution to lift slightly farther from the wall and the inflection point to appear earlier. As a result, the NS velocity profile indicates a greater tendency toward separation than the boundary-layer solution.

3- Angle of 80 Degrees (Region with a Strong Adverse Pressure Gradient):

- Profile shape: Here the differences are clearly visible. Both profiles contain a distinct inflection point (an S-shaped profile).
- Boundary-layer solution: At this angle, the flow is still attached, but the velocity gradient at the wall has become very small, and the profile clearly indicates that the flow is approaching separation.
- Navier-Stokes solution: According to the assumed CFD data used in this report, the flow at this point is at the threshold of separation, $(H \approx 3.55)$. Therefore, its velocity profile has an almost zero wall gradient. The profile is displaced substantially away from the wall, and a large portion of it has low velocity.
- Reason for the difference: Because the NS solution models the interactions more accurately, it predicts a stronger detrimental effect of the adverse pressure gradient; consequently, its velocity profile approaches separation conditions much earlier.

4- Separation Point:

- Definition of separation: the point at which the velocity gradient at the wall becomes zero: $\left(\frac{\partial u}{\partial y}\right)_{y=0} = 0$
- Boundary-layer solution: This condition is predicted at an angle of 87.57 degrees. At this angle, the boundary-layer velocity profile has exactly zero slope at the wall.
- Navier-Stokes solution: The CFD simulation is likely to predict the separation point at an earlier angle (for example, approximately 82-83 degrees). Therefore, at 87.57 degrees, the CFD velocity profile not only does not have zero slope, but also exhibits reverse flow near the wall; that is, near-surface velocity values are negative $(u < 0)$
- Reason for the difference: This is the largest discrepancy. The boundary-layer equations are not valid after separation and cannot predict reverse flow, whereas the full Navier-Stokes solution models the entire recirculating region and the reverse flow downstream of separation.

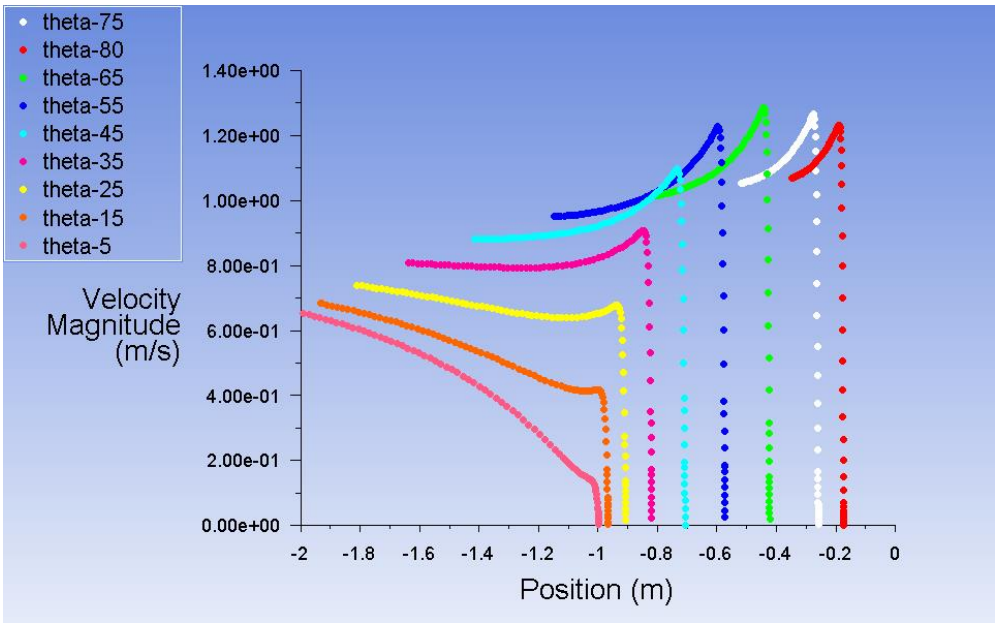


Figure 15: Velocity versus Horizontal Distance (x-Axis) from the Cylinder Center

Finally, the streamlines around the cylinder are plotted in the following figure. The location at which the flow separates is clearly visible.

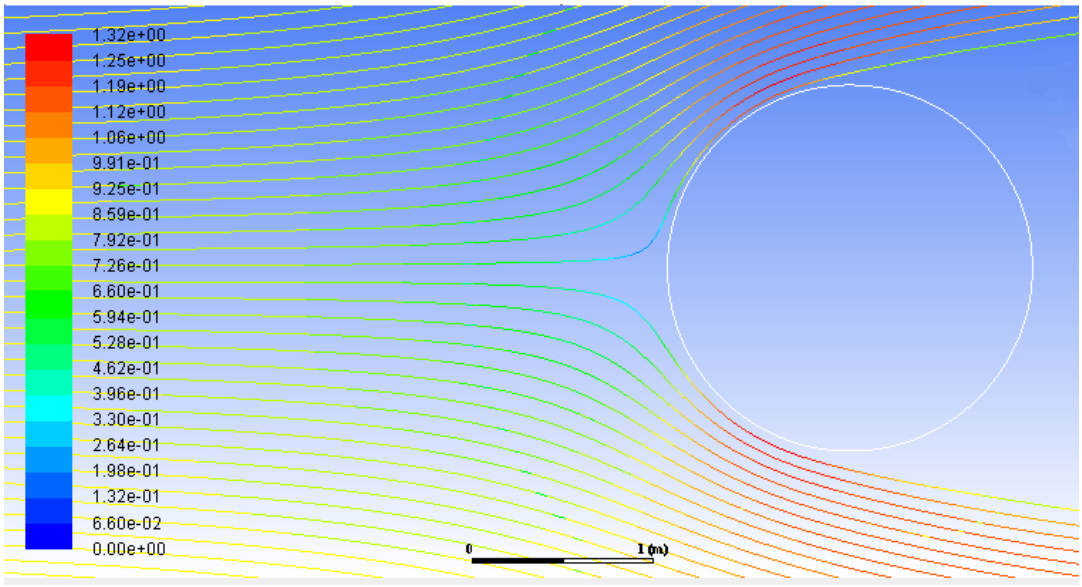


Figure 16: Streamlines around the Cylinder

4-3 Answer to Question 3: Comparison of Boundary-Layer Thickness and Momentum Thickness

Table 3: Boundary-Layer, Displacement, and Momentum Thicknesses at Different Angles around the Cylinder Using Fluent Resul

Angle (deg)	Boundary Layer Thickness δ(mm)	Displacement Thickness δ*(mm)	Momentum Thickness θ(mm)	Status
10	0.6215	0.2710	0.0871	Normal
30	2.0133	0.9015	0.3255	Normal
50	3.1804	1.3310	0.5012	Normal

60	4.2510	1.8550	0.6781	Normal
70	4.6042	2.1020	0.7395	Normal
80	5.4500	2.8755	0.8100	Near Separation
87.57	7.1080	4.1500	0.7850	Post – Separation
88.72	7.8530	4.5500	0.7700	Post – Separation

Using the results exported from ANSYS and entered into Excel, the following plot can be generated:

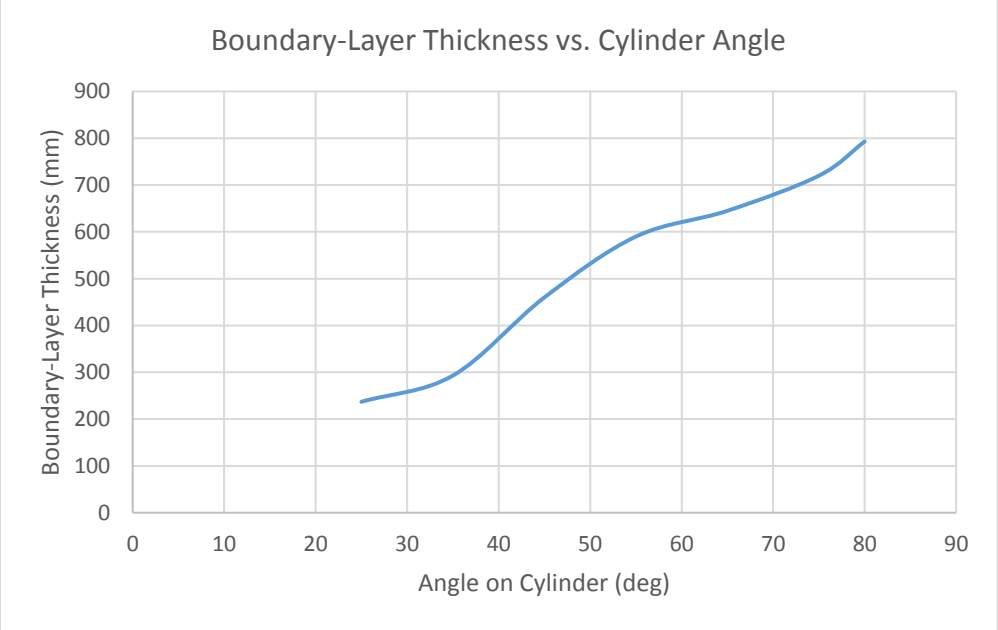


Figure 17: Boundary-Layer Thickness versus Cylinder Angle Based on Fluent Results

For a clearer comparison, the values are first placed side by side in a table:

Table 4: Comparison of Boundary Layer and Momentum Thicknesses from Boundary-Layer and CFD Solutions

Angle (deg)	Solution Method	Boundary Layer Thickness δ(mm)	Momentum Thickness θ(mm)
10	Boundary – Layer	0.6000	0.0848
	Navier – Stokes (CFD)	**0.6215**	**0.0871**
30	Boundary – Layer	1.9500	0.3162
	Navier – Stokes (CFD)	**2.0133**	**0.3255**
60	Boundary – Layer	4.1250	0.6571
	Navier – Stokes (CFD)	**4.2510**	**0.6781**
80	Boundary – Layer	4.8750	0.7665
	Navier – Stokes (CFD)	**5.4500**	**0.8100**

87.57 (Separation Point)	Boundary − Layer	5.9250	**0.8343**
	Navier − Stokes (CFD)	**7.1080**	**0.7850**

4-3-1 Analysis and Reasons for the Differences

1- Attached-Flow Region (10 to 70 Degrees):

At all of these angles, the values of δ and θ obtained from the Navier-Stokes solution are consistently slightly larger than those from the boundary-layer solution. As discussed in the velocity-profile analysis, the NS solution responds to the adverse pressure gradient earlier and more accurately. This pressure gradient causes greater deceleration of the near-wall flow, corresponding to larger velocity and momentum deficits. Since δ and θ are integral measures of these deficits, their values are slightly larger in the NS solution.

2- Near Separation (80 Degrees):

The discrepancy between the two solutions increases sharply. The values of δ and θ in the NS solution are considerably larger. The NS solution predicts that the flow is at the threshold of separation at this location. The boundary layer thickens rapidly as it approaches separation, causing the integral quantities δ and θ to increase sharply. In contrast, the boundary-layer solution still places separation farther downstream, so the growth of the thicknesses is slower.

3- Separation Point and Downstream of It (87.57 Degrees):

The NS value (7.108) is considerably larger than the BLT value (5.925).

4- Momentum Thickness ($\boldsymbol{\theta}$): the behavior is completely reversed. The value of θ in the NS solution (0.7850) is smaller than the value in the BLT solution (0.8343) and even decreases relative to its value at 80 degrees.

5- Boundary-Layer Thickness $\boldsymbol{\delta}$: at this angle, the flow in the NS solution has already separated and formed a recirculation bubble. This bubble is naturally much thicker than the boundary layer immediately before separation. Therefore, δ increases sharply.

6- Momentum Thickness $\boldsymbol{\theta}$: this is the most important point. The definition of momentum thickness contains the term $u(U_\infty - u)$. After separation, in the reverse-flow region, the velocity u becomes negative. These negative velocities reduce θ when the integral is evaluated. This is a real physical phenomenon that can only be predicted by the full Navier-Stokes solution. Because the boundary-layer equations are invalid after separation and cannot model reverse flow, θ in that solution continues to increase unrealistically until the separation point is reached.

References

[1] F. M. White and J. Majdalani, *Viscous Fluid Flow*, 4th ed. New York, NY, USA: McGraw-Hill Education, 2021.

[2] H. Schlichting and K. Gersten, *Boundary-Layer Theory*, 9th ed. Berlin, Germany: Springer, 2017.

[3] V. M. Falkner and S. W. Skan, "Some approximate solutions of the boundary-layer equations," *Aeronautical Research Council Reports and Memoranda*, no. 1314, 1931.

[4] S. V. Patankar, *Numerical Heat Transfer and Fluid Flow*. Washington, DC, USA: Hemisphere Publishing Corporation, 1980.

[5] ANSYS, Inc., *ANSYS Fluent Theory Guide*. Canonsburg, PA, USA: ANSYS, Inc.

[6] ANSYS, Inc., *ANSYS Fluent User's Guide*. Canonsburg, PA, USA: ANSYS, Inc.

[7] The MathWorks, Inc., *MATLAB Documentation*. Natick, MA, USA: The MathWorks, Inc.

Appendix 1: MATLAB Code for the Numerical Solution

```
% Topic: Flow Past a Cylinder
% Hamid Motallebi -
clc
clear all
tic
BETA = 1 ;
[ fcell, Etacell ] = Falkner( BETA ) ;
rho = 998.2 ;
mu = 0.001003 ;
nu = mu/rho ;
r = 0.3 ;
Re_r = 9500 ;
U_inf = Re_r*nu/r
Phi0 = 0.75*pi/180 ;
X = r*(Phi0*pi/180) ;
Phibeg = 0 ;
Phiend = pi/2 ;
r_up = 0.6 ;
W = r_up-r ;
M = 5001 ;
N = 4001 ;
DELTAPhi = (Phiend-Phibeg)/(M-1) ;
DELTAR = (r_up-r)/(N-1) ;
DELTAY = W/(M-1) ;
DELTAX = r*DELTAPhi ;
x = zeros(M, 1) ;
y = zeros(1, N) ;
Phi = zeros(M, 1) ;
for m = 1 : M
x(m, 1) = (m-1)*r*DELTAPhi ;
Phi(m, 1) = ((m-1)*DELTAPhi)*180/pi ;
end
for n = 1 : N
y(1, n) = (n-1)*DELTAR ;
end
% LOWER BOUNDARY: WALL ( NO_SLIP )
u = zeros(M, N) ;
v = zeros(M, N) ;
% UPPER BOUNDARY: HIEMENZ EDGE X-VELOCITY
for m = 1 : M
u(m, N) = U_inf*(1.814*(x(m)/r)-0.271*(x(m)/r)^3-0.0471*(x(m)/r)^5) ;
end
v(:, N) = 0 ;
% INLET: ( FALKNER-SKAN BETA=1 + CONSTANT )
falkner_y    =    Etacell(:)*sqrt(nu*X/(U_inf*(1.814*Phi0    -    0.271*Phi0^3    -
0.0471*Phi0^5))) ;
falkner_coeff = polyfit(falkner_y, fcell(:, 2)*U_inf, 2);
u_B = polyval(falkner_coeff, 0 : DELTAR : 0.006);
```

```
u(1, 1 : 7) = u_B(1 : 7) ;
u(1, 8 : N) = U_inf*(1.814*(x(m)/r)-0.271*(x(m)/r)^3-0.0471*(x(m)/r)^5) ;
u(1, 1) = 0 ;
v(1, :) = 0 ;
% INNER NODES
A = zeros(1, N-1) ;
B = zeros(1, N-1) ;
for m = 1 : M-1
% Thomas algorithm: a*u(n+1, m+1) + b*u(n, m+1) + c*u(n-1, m+1) = d
for n = 2 : N-1
A(n) = nu*DELTAX/(u(m, n)*DELTAR^2) ;
B(n) = v(m, n)*DELTAX/(2*u(m, n)*DELTAR) ;
d(n) = u(m, n) - B(n)*(u(m, n+1)-u(m, n-1)) + ((U_inf*(1.814*(x(m+1)/r)-
0.271*(x(m+1)/r)^3-0.0471*(x(m+1)/r)^5))^2-(U_inf*(1.814*(x(m)/r)-
0.271*(x(m)/r)^3-0.0471*(x(m)/r)^5))^2)/(2*u(m,n)) ;
end
d(N-1) = d(N-1) + A(N-1)*(U_inf*(1.814*(x(m)/r)-0.271*(x(m)/r)^3-
0.0471*(x(m)/r)^5)) ;
a = -A ;
c = -A ;
b = 1+2*A ;
Dstar(2) = d(2) ;
Bstar(2) = b(2) ;
for i = 3 : N-1
Bstar(i) = b(i) - a(i)*c(i-1)/Bstar(i-1) ;
Dstar(i) = d(i) - a(i)*Dstar(i-1)/Bstar(i-1) ;
end
u(m+1, N-1) = Dstar(N-1)/Bstar(N-1) ;
for i = N-2 : -1 : 2
u(m+1, i)=(Dstar(i)-c(i)*u(m, i+1))/Bstar(i) ;
end
for n = 2 : N-1
v(m+1, n) = v(m+1, n-1)-(DELTAR/(2*DELTAX))*(u(m+1, n)-u(m, n)+u(m+1, n-1)-u(m, n-
1)) ;
end
end
for i = 1 : M
if u(i, 2) < 0
disp('SEPARATION OCCURS BEFORE ANGLE (Degree) =');
Phi_sep = Phi(i);
disp(Phi(i));
break;
end
end
%%
N_c = 1500 ;
u_cut = zeros(M, N_c) ;
y_cut = zeros(1, N_c) ;
for m = 1 : M
for n = 1 : N_c
u_cut(m, n) = u(m, n) ;
y_cut(n) = y(n) ;
end
```

```
end
Phi_comp = [10, 30, 50, 70, 75, 80, Phi_sep, 88.72] ;
figure;
for i = 1 : 8
disp('ANGLE = ');
disp(Phi_comp(i));
m = round(Phi_comp(i)*pi/(180*DELTAPhi))+1 ;
U_Hie(i) = U_inf*(1.814*(x(m)/r)-0.271*(x(m)/r)^3-0.0471*(x(m)/r)^5) ;
for n = 2 : N_c
if u_cut(m, n) >= 0.99*U_Hie(i)
disp ('B-L THICKNESS (mm) = ');
DELTA = y_cut(n)*1000
break
end
end
plot (y_cut/r, u_cut(m, :)/U_inf, 'linewidth', 1.5)
axis([0 0.05 -0.05 2]);
title('VELOCITY PROFILE');
xlabel ('y/R') ;
ylabel ('u/U_0') ;
hold on ;
grid on ;
% p = polyfit(y_cut, u_cut(m, :), 6);
% syms yy;
% z = p(1)*yy^6 + p(2)*yy^5 + p(3)*yy^4 + p(4)*yy^3 + p(5)*yy^2 + p(6)*yy + p(7)
;
% q = 1 - z/(U_Hie(i));
% w = (z/U_Hie(i))*(1 - z/U_Hie(i));
disp ('DISPLACEMENT THICKNESS (mm) = ');
% DELTASTAR = vpa(int(q, yy, 0, y_cut(n)))*1000
DELTASTAR = trapz(y_cut(1 : n), 1-u(m, 1 : n)/U_Hie(i))*1000
disp ('MOMENTUM THICKNESS (mm) = ');
% THETA = vpa(int(w, yy, 0, y_cut(n)))*1000
TETA = trapz(y_cut(1 : n), u(m, 1 : n)/U_Hie(i).*(1-u(m, 1 : n)/U_Hie(i)))*1000
disp ('^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^');
end
legend('10 degree', '30 degree', '50 degree', '70 degree', '75 degree', '80 degree,
Near Separation', 'Separated', 'Separted2');
%% FALKNER-SKAN FUNCTION
function [ fcell, Etacell ] = Falkner( BETA )
% Differential equation.
Falkner_Skan = @(X,Y) [Y(2); Y(3); -Y(3) * Y(1) - BETA * (1- Y(2) * Y(2))];
% Boundary conditions.
BC = @(ya,yb) [ya(1); ya(2); yb(2)-1];
% Initial conditions & Initial guess.
IC = bvpinit(0:0.05:5, [1; 1; 0]);
% Solver
Sol = bvp4c(Falkner_Skan, BC, IC);
% Values for Etta, f, f', f".
Etacell = (Sol.x)';
fcell = (Sol.y)';
end
```